



Technical Service Bulletin

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New Water Pump and Service Kits for K19 and QSK19

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Introduction

This document introduces new water pump and service kits for K19 and QSK19 diesel engines. The new water pump is used in applications that operate in high levels of atmospheric contamination. The new service kits improve water pump durability by reducing the possibility of environmental contamination through the addition of a weep hole vent tube.

The rail application water pumps have also been updated with a new controlled dimension weep hole to accommodate the vent tube and fitting. The orientation of the weep hole was moved on the Rail Application water pump. The new part number, for rail application **only**, is in Table 1.

Table 1- New Water Pump (Rail Application Only)

Description	Part Number
Pump, Water	4998217

To replace the water pump on a QSK19 rail engine, a new kit, Part Number 2881826, has been released. The kit contains all of the parts listed in Table 2.

Table 2- Water Pump Kit, Part Number 2881826

Description	Part Number	Quantity
Seal, O-ring	145623	1
Gasket, Water Transfer Connection	205288	1
Gasket, Water Transfer Connection	205289	1
Seal, O-ring	3028291	2

Table 2- Water Pump Kit, Part Number 2881826

Description	Part Number	Quantity
Gasket, Connection	3060912	1
Gasket, Water Pump	3202283	1
Pump, Water	4998217	1
Hose, Plain	2888750	1
Fitting, Vent	4998211	1
Screw, Hex Flange Head Cap	3910665	1
Clip	4992548	1

If installing the new improved water pump on an Industrial engine, order the following parts listed in Table 3:

Table 3- Parts to Upfit Industrial Pump

Description	Part Number	Quantity
Hose, Plain	2888751	1
Fitting, Vent	4998211	1
Clip	4992548	1

The water pump part number did **not** change. However, these parts were added to the following options on 19-Sep-2011, with engine serial number (ESN) first 37251183.

Upgraded Options

- WP4010
- WP4031
- WP4032
- WP4702
- WP4706
- WP4712
- WP4713
- WP4714

Assembly instructions are outlined below.

Railcar Upgrade Kit Instruction

This upfit requires a replacement of the water pump assembly, from Part Number 4081093 to Part Number 4998217.

Coat the first two barbs of the 90° fitting, Part Number 4998211, with medium strength silicone sealant, Part Number 3823494, or equivalent. Drive it into the weep hole of the pump until the shank touches the pump housing.

Do **not** put sealant on the tip of the barbed fitting, as this could cause contamination of the pump and/or plugging of the fitting upon installation.

Once the fitting is installed in the weep hole, install the hose, Part Number 4998213, by hand, over the fitting until the end of the hose touches the shank of the fitting.

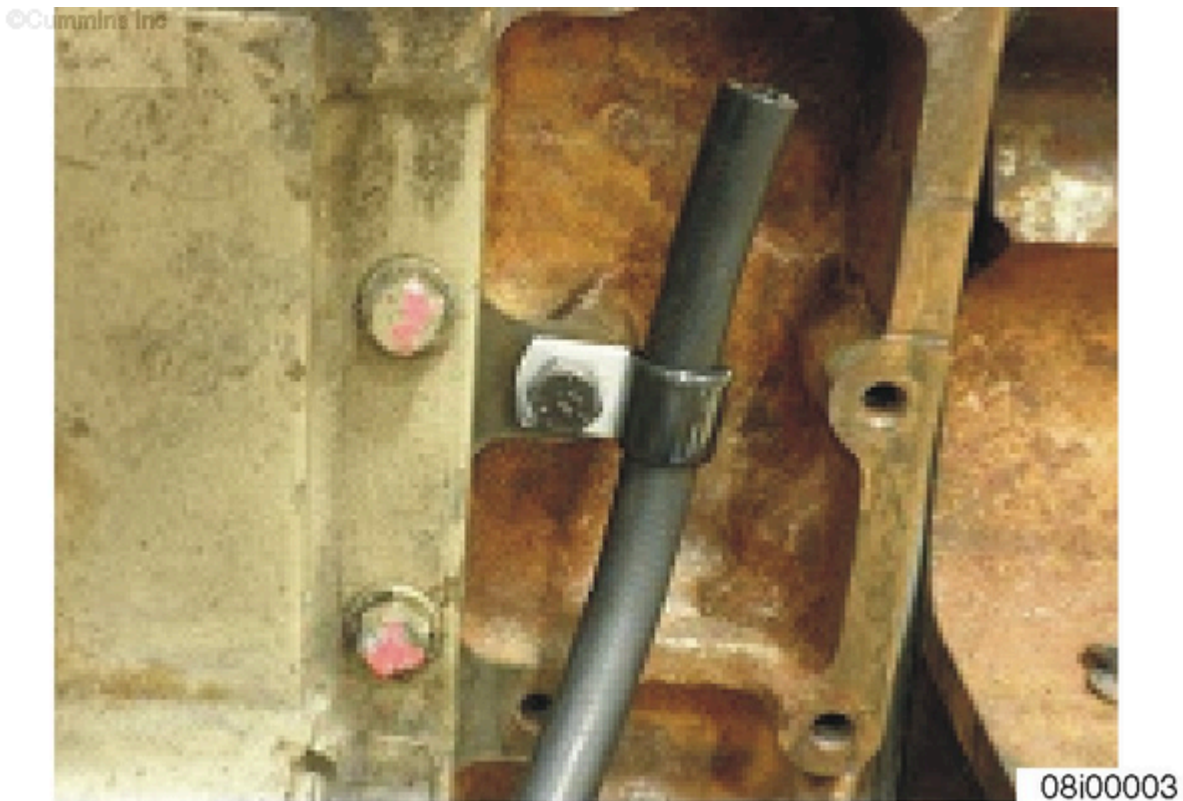
Use the following procedure in the Service Manual, QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS, Bulletin 4021592, to install the water pump. Refer to Procedure 008-062 in Section 8. ([/qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html](http://qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html))

Slide the p-clip, Part Number 4992548, over the loose end of the hose. Align the bolt hole of the p-clip with the tapped boss at the lower right corner of the oil cooler housing.

Secure the hose and p-clip to the block using capscrew, Part Number 3910665. Tighten the capscrew.

Torque Value 45 N•m [33 ft-lb]

Be sure **not** to twist or kink the hose.



If necessary, cut any extra length of hose that would potentially be submerged from lying too close to the block.

Industrial Pump Upgrade Kit Instructions

This upfit can be performed on new water pumps **only** to protect the shaft from atmospheric contamination.

Coat the first two barbs of the 90° fitting, Part Number 4998211, with medium strength silicone sealant, Part Number 3823494, or equivalent. Drive the barb into the weep hole of the water pump until the shank touches the pump housing.

Do **not** put sealant on the tip of the barbed fitting, as this could cause contamination of the pump and/or plugging of the fitting upon installation.

Once the fitting is installed in the weep hole, install the hose, Part Number 4998213, by hand over the fitting, until the end of the hose touches the shank of the fitting.

Use the following procedure in the Service Manual, QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS, Bulletin 4021592, to install the water pump. Refer to Procedure 008-062 in Section 8. ([/qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html](http://qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html))

Secure the hose and p-clip, Part Number 4992548, to the hand hole cover, using the hand hole cover capscrew.

Torque Value 45 N•m [33 ft-lb]

Be sure **not** to twist or kink the hose. The hose **must** go continuously down from the weep hole. Its lowest point **must** be its open end.



If necessary, cut the hose to make sure its open end is unobstructed and in clear space.

Document History

Date	Details
2011-12-1	Module Created

Last Modified: 20-Dec-2011
